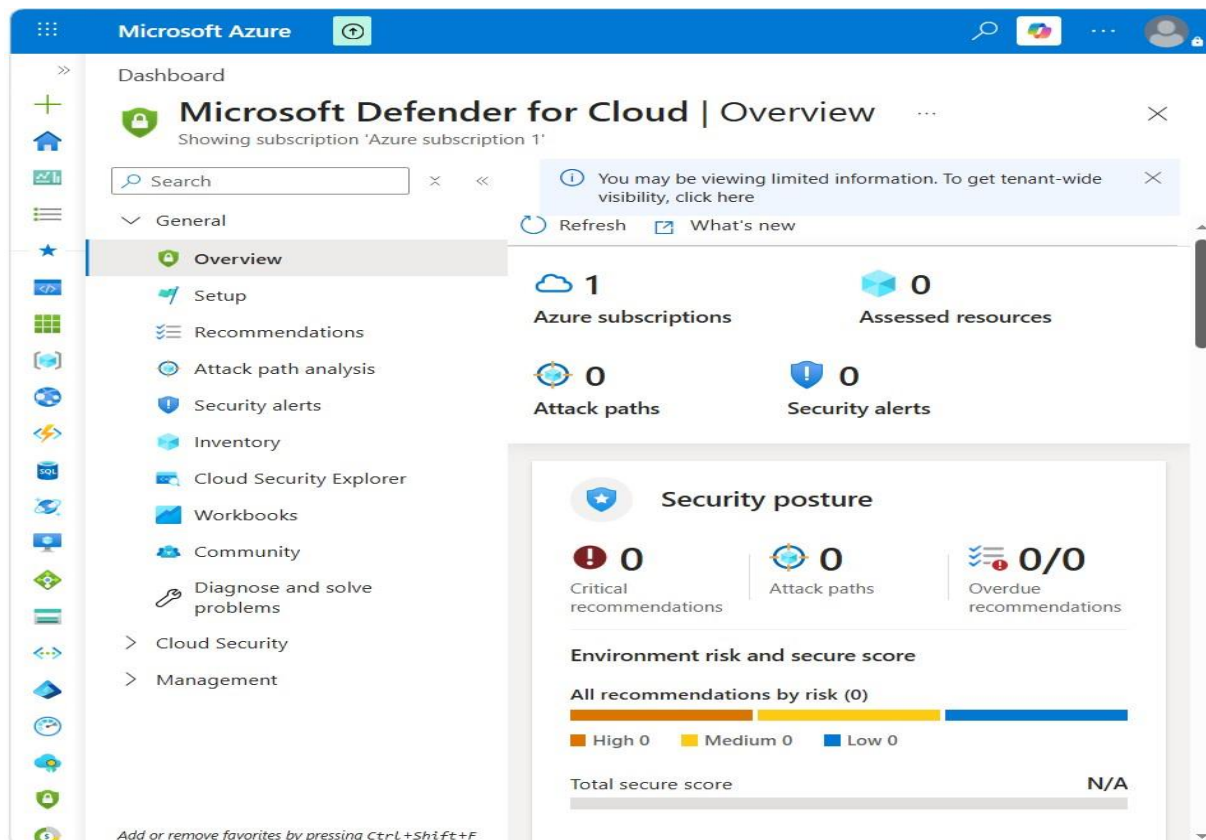


Task 1: Security Posture Assessment

Before having to make any changes, you must capture the "vulnerable" state of your Virtual Machines environment.

1. **Access Defender:** In the Azure Portal search bar, type **Microsoft Defender for Cloud** which will show off certain scores.
2. **Locate to Secure Score:** On the **Overview** blade, look for the **Secure Score** tile.
3. **Identify Recommendations:** Click on **Recommendations** in the left-hand menu.
 - o *Action:* Filter by **Status: Active**. Identify the top three recommendations.
 - o *Note:* Look for high-severity items like "Management ports which should be closed" or "System updates should be installed."
4. **Analyze Risk:** Click into each of the top three recommendations to read the **Description** and **Remediation steps**.



Task 2: Remediation

2a: Close Open Management Ports

This task involves modifying your **Network Security Group (NSG)**.

1. **Find the NSG:** Navigate to your Virtual Machine. Under the **Settings** section, click **Networking**.
2. **Identify the Risk:** Look for Inbound Port Rules allowing **TCP 3389 (RDP)** or **TCP 22 (SSH)** with a Source of "Any" or "Internet."
3. **Remediate:**
 - Its Either we **Delete** the rule entirely.
 - Or **Edit** the rule to change the "Source" from "Any" to "IP Addresses" and input your specific public IP to work effectively.
4. **Confirm:** Ensure the "DenyAllInBound" rule is now the primary gatekeeper for those ports.

2b: Enable OS Updates

We use **Azure Update Manager** to ensure our VM stays patched.

1. **Navigate:** Search for **Update Manager** in the portal.
2. **Check Settings:** Go to **Machines** on the left, select your VM, and look at the "Update settings."
3. **Configure:**
 - Click **Update settings** -> **Change settings**.
 - Set **Patch orchestration** to "Automatic by platform" (for Azure orchestration) or "Image default."
 - Enable **Periodic Assessment**
4. **Save:** Once saved, trigger a "Check for updates" to show the system is actively communicating with the Update Manager.

Microsoft Azure | Upgrade

Home > Azure Update Manager

Azure Update Manager | Machines

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1 out of 1 machine(s) don't have periodic assessment enabled. Enable now for supported machines to regularly check for updates. [Enable now](#) →

Filter by name

Subscription: Azure subscription 1 | Resource group: techbridge-rg | Resource type: All | Workloads: All

Location: South Africa North | OS: All | Update classifications: All | Patch orchestration: All

Periodic assessment: All | Hotpatch status: All | Status: All | ESU status: All | Tags: All

Total machine(s) 1

No updates data 1

No pending updates 0

Pending updates 0

Pending Reboot 0

Unsupported 0

Showing 1 of 1 records

No grouping

Name	Update status	ESU Status	Operating system	Resource type	Orchestration M...
Techbridge-VM	No updates data	N/A	Linux	Azure virtual machine	Not Applicable

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Total machine(s) 1

No updates data 1

No pending updates 0

Pending updates 0

Pending Reboot 0

Unsupported 0

Showing 1 of 1 records

No grouping

Name	Update status	ESU
Techbridge-VM	No updates data	N/A

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